

Zachary Ryan

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EDUCATION

Robert Morris University

Bachelor of Science in Cybersecurity

Outstanding Cybersecurity Student Award [\[Link\]](#) | Dean's List Recipient

Pittsburgh, PA

August 2024 - December 2026

GPA: 3.92/4.0

EXPERIENCE

O.F. Mossberg & Sons, Inc. | Artificial Intelligence Intern | North Haven, CT

August 2026 - Present

- Delivered company-wide AI training for executives and cross-functional teams on safe, responsible AI use and best practices
- Automated Concur export workflows for credit-card and reimbursement pivots, reducing 5-10 monthly hours to minutes
- Developed a controlled AI-assisted workflow to analyze three email sources, identify outstanding payments, and automate missed/late payment reviews, replacing hours of manual searching with a repeatable process that returns results in minutes

Brixmor Property Group | Cybersecurity Intern | Philadelphia, PA

May 2026 - July 2026

- Triaged security tickets across a 500+ user enterprise environment using enterprise security tools and platforms
- Performed Rapid7 vulnerability management on 25+ high-risk assets, documenting CVEs and remediation recommendations
- Assessed 100+ Cisco Umbrella domains, classified DNS risk, and recommended block/allow decisions
- Assisted with Third-Party Risk Management policy development for vendor onboarding, technology review, security documentation, control assessment, risk acceptance, compliance review, and vendor security evaluation workflows

United States Navy | Machinist Mate - Nuclear Reactor Department | Norfolk, VA

July 2020 - September 2023

- Monitored reactor support systems, logs, gauges, alarms, and indicators to detect abnormal operating conditions
- Diagnosed mechanical and reactor system abnormalities using technical manuals, schematics, and procedures
- Executed incident response procedures under strict safety, documentation, and operational requirements

PROJECTS

ReactorBench-LM | From-Scratch Transformer Research (In Progress) [\[GitHub\]](#)

- Built a from-scratch decoder-only PyTorch Transformer for fictional event narratives without pretrained weights or real data
- Built reproducible data, tokenizer, training, and evaluation pipelines with deterministic tests and Safetensors checkpoints
- Implemented held-out evaluation for structured outputs, using results to guide a documented remediation plan

AI-Powered Phishing Email Detection & Safe Rewrite System [\[GitHub\]](#)

- Built a full-stack phishing detection platform using FastAPI and React to classify emails by risk level
- Implemented explainable risk scoring using keyword analysis, links, and SPF/DKIM/DMARC authentication headers
- Developed a safe email rewrite engine that removes malicious content and highlights changes using HTML diff visualization

Policy-as-Code Docker Network Segmentation Lab [\[GitHub\]](#)

- Designed a segmented enterprise network architecture with public, private, and management security zones
- Implemented policy-as-code controls and firewall rules to restrict unauthorized network access
- Developed automated testing and visualization tools to validate security policy compliance across network segments

PUBLICATIONS

Ryan, Z., & Mishra, S. (2026). *Auditable Evidence in AI Cybersecurity Compliance: A Cross-Framework Analysis*.

Computers & Security (Submitted June 2026)

- Compared NIST AI RMF, ISO/IEC 42001, EU AI Act, CSA AICM, AWS, and Google Cloud security guidance
- Analyzed evidence gaps across documentation, logging, monitoring, access control, model security, and testing
- Evaluated how logs, access records, monitoring outputs, and assessment artifacts support cybersecurity compliance
- Identified gaps between AI security requirements, audit evidence, and assurance readiness across major frameworks

Ryan, Z., & Mishra, S. (2026). *Evaluating AI vs Human Phishing Evasion Across Emulated Email Filters and SpamAssassin*.

NEDSI Annual Conference (Accepted; presented April 2026) [\[Publication Link - PDF pp. 189- 202\]](#)

- Created controlled prompts to generate realistic AI phishing emails using dummy URLs and fictional identities
- Built a 1,260-email dataset comparing AI and human phishing across account, financial, and delivery themes
- Applied Chi-square and Z-tests showing AI phishing evaded selected emulated filters more often than human phishing emails
- Developed a Python-based filter emulation framework to score phishing emails, compare AI versus human bypass rates, evaluate provider-specific thresholds, and test emulated Gmail, Outlook, Yahoo, and SpamAssassin detection conditions

Skills

Languages & AI: Python | SQL | Bash | PowerShell | Java | TypeScript | PyTorch | SentencePiece | Safetensors

IT & Security Tools: CrowdStrike Falcon | Rapid7 | Azure | Mimecast | Okta | | FreshService | Microsoft Defender | Cisco Umbrella

Technology & Development: Linux | Docker | Git/GitHub | TCP/IP | Firewalls | Network Segmentation | FastAPI | React

Security Operations: Vulnerability Management | Incident Response | Security Investigations | Ticket Triage | Access Control

Governance & Risk: TPRM | ISO/IEC 42001 | NIST AI RMF | Cyber Policy | Audit Evidence | Risk Management

LICENSES AND CERTIFICATIONS

- DoD Secret Clearance | CompTIA Security+ (Scheduled) | Google Cybersecurity Professional Certificate [\[Link\]](#)